

# Assignment 1 --CDC Diabetes Health Indicator

## EDA & CLASSIFICATION REPORT

KARTHIKEYAN R – BL.SC.P2DSC26034

**Dataset shape:** 253680 rows  $\times$  22 columns (**source:** CDC BRFSS 2015, UCI ML Repository id=891). KNN/Naive Bayes trained on a **stratified 15,000-row sample**

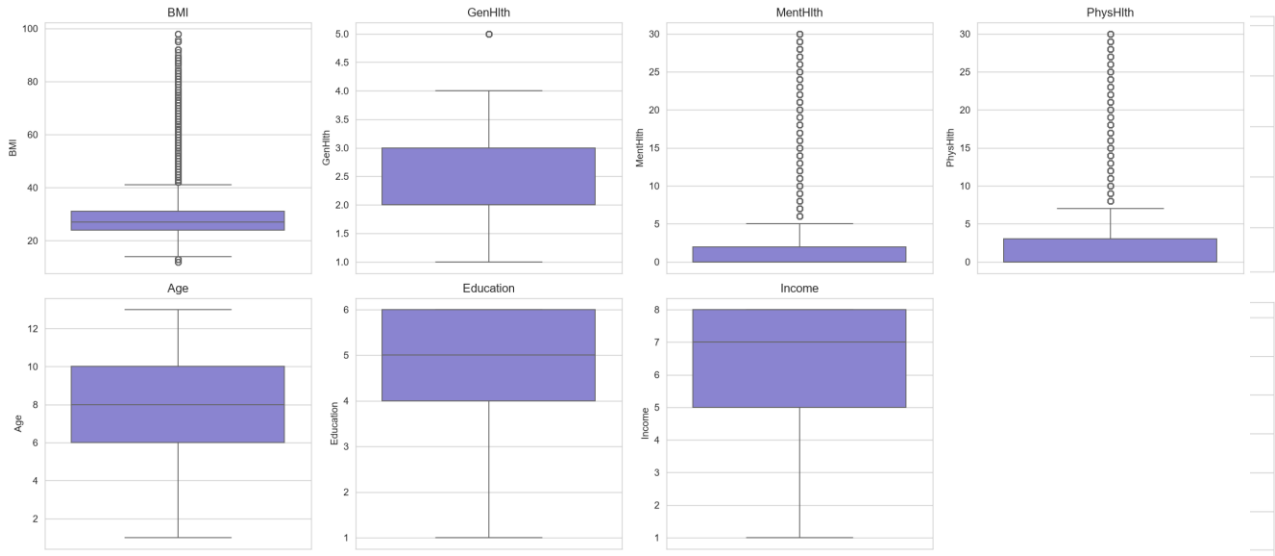
## 1. Data Quality Report

Missing values were imputed using class-conditional medians before this report was generated.

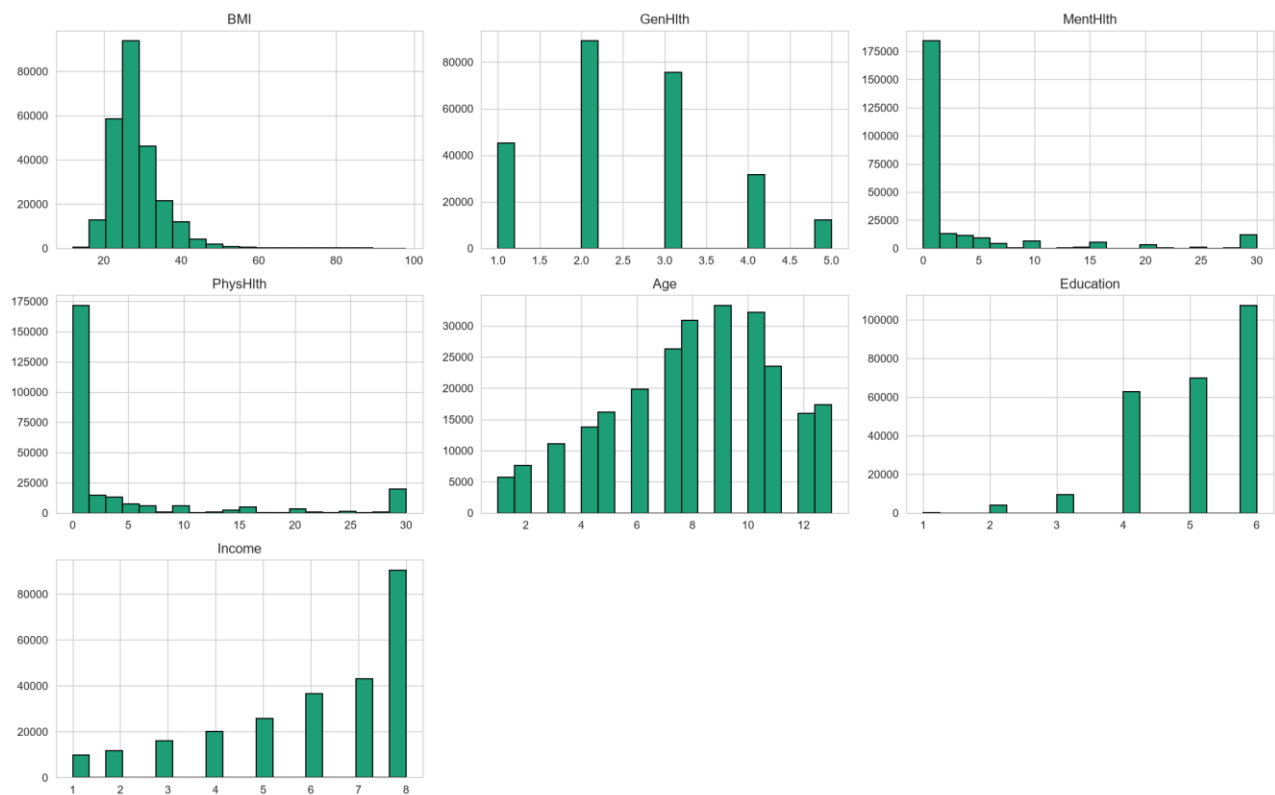
|                      | dtype | cardinality |
|----------------------|-------|-------------|
| HighBP               | int64 | 2           |
| HighChol             | int64 | 2           |
| CholCheck            | int64 | 2           |
| BMI                  | int64 | 84          |
| Smoker               | int64 | 2           |
| Stroke               | int64 | 2           |
| HeartDiseaseorAttack | int64 | 2           |
| PhysActivity         | int64 | 2           |
| Fruits               | int64 | 2           |
| Veggies              | int64 | 2           |
| HvyAlcoholConsump    | int64 | 2           |
| AnyHealthcare        | int64 | 2           |
| NoDocbcCost          | int64 | 2           |
| GenHlth              | int64 | 5           |
| MentHlth             | int64 | 31          |
| PhysHlth             | int64 | 31          |
| DiffWalk             | int64 | 2           |
| Sex                  | int64 | 2           |
| Age                  | int64 | 13          |
| Education            | int64 | 6           |
| Income               | int64 | 8           |
| Outcome              | int64 | 2           |

## 2. Visualizations

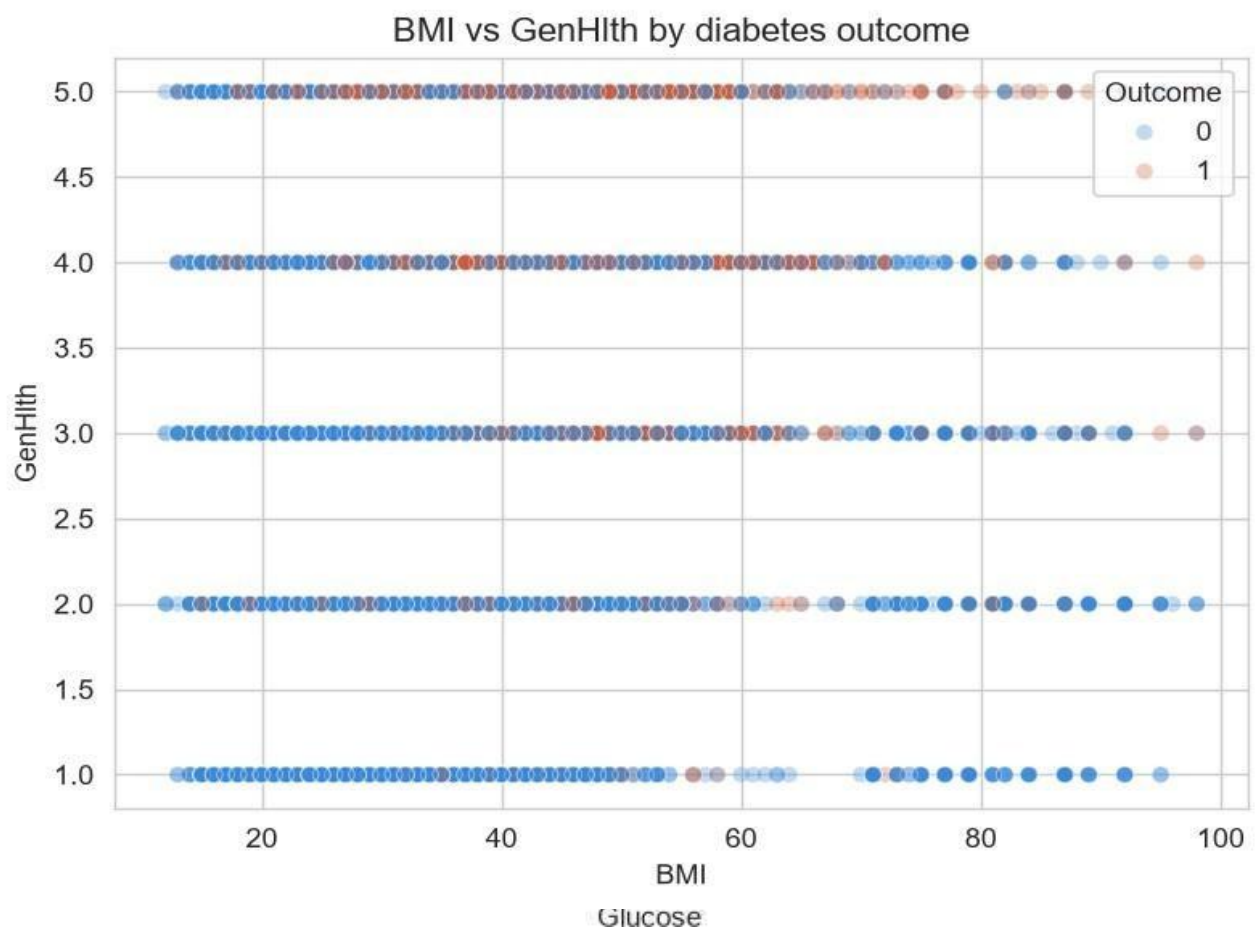
### Boxplots



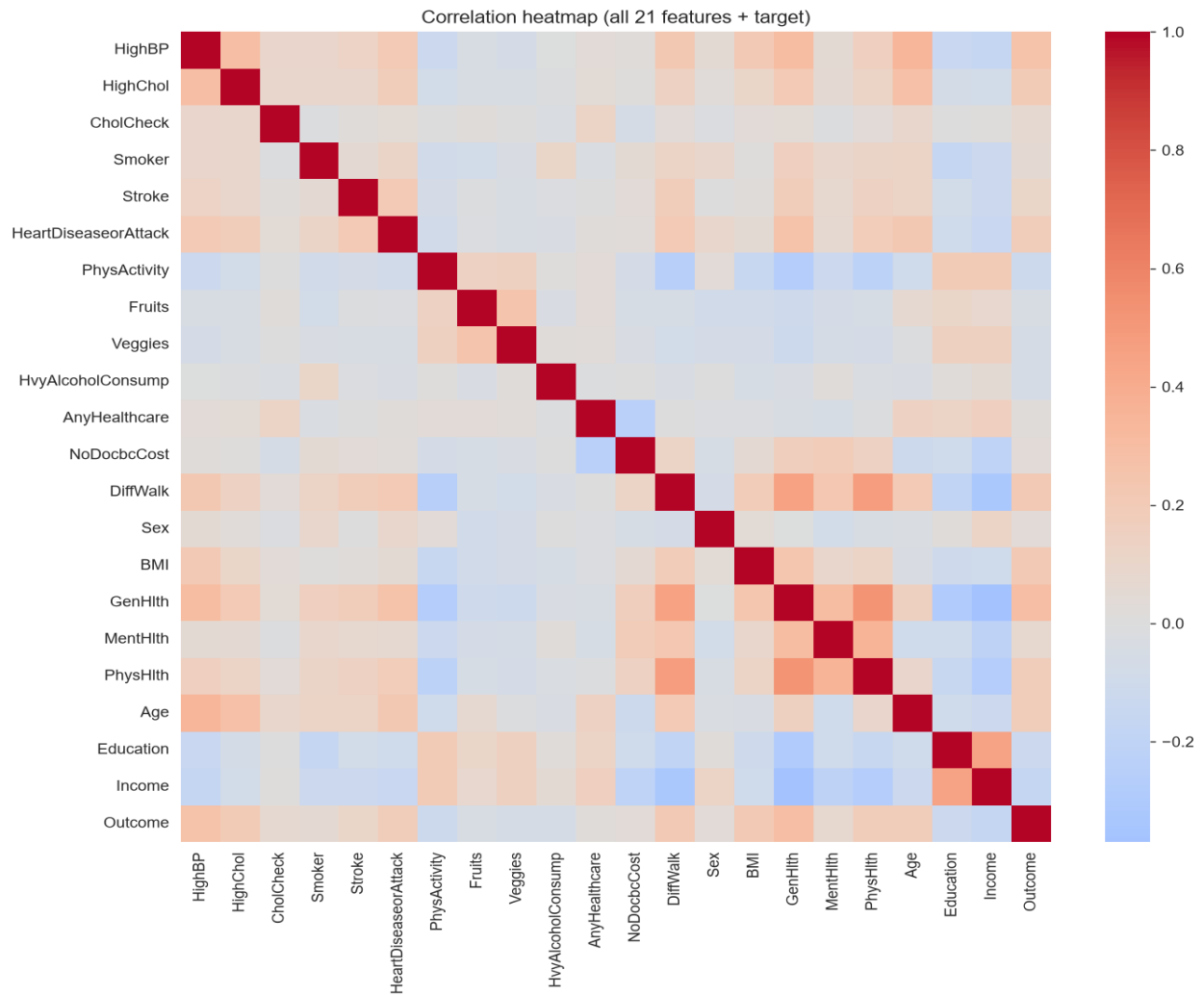
### Histograms



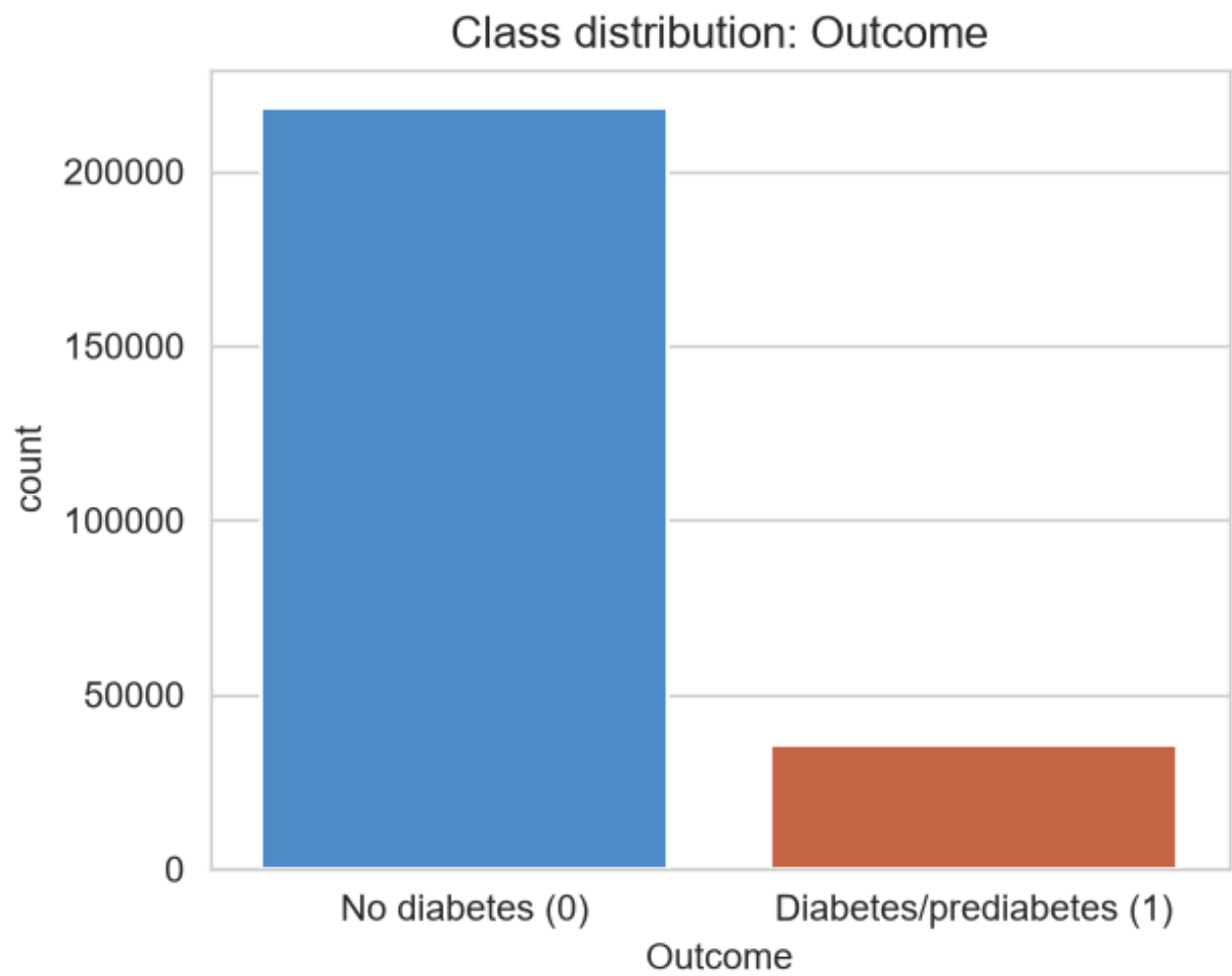
## Scatter: Glucose vs BMI



# Correlation heatmap



## Class Distribution



### 3. Model Results (*KNN vs Weighted KNN vs Naive Bayes*)

| Model                               | Accuracy | Precision | Recall | F1-score | ROC-AUC |
|-------------------------------------|----------|-----------|--------|----------|---------|
| <b>KNN (k=5, uniform)</b>           | 0.704    | 0.261     | 0.612  | 0.366    | 0.709   |
| <b>Naive Bayes</b>                  | 0.712    | 0.290     | 0.734  | 0.416    | 0.781   |
| <b>Weighted KNN (k=5, distance)</b> | 0.711    | 0.265     | 0.605  | 0.369    | 0.710   |

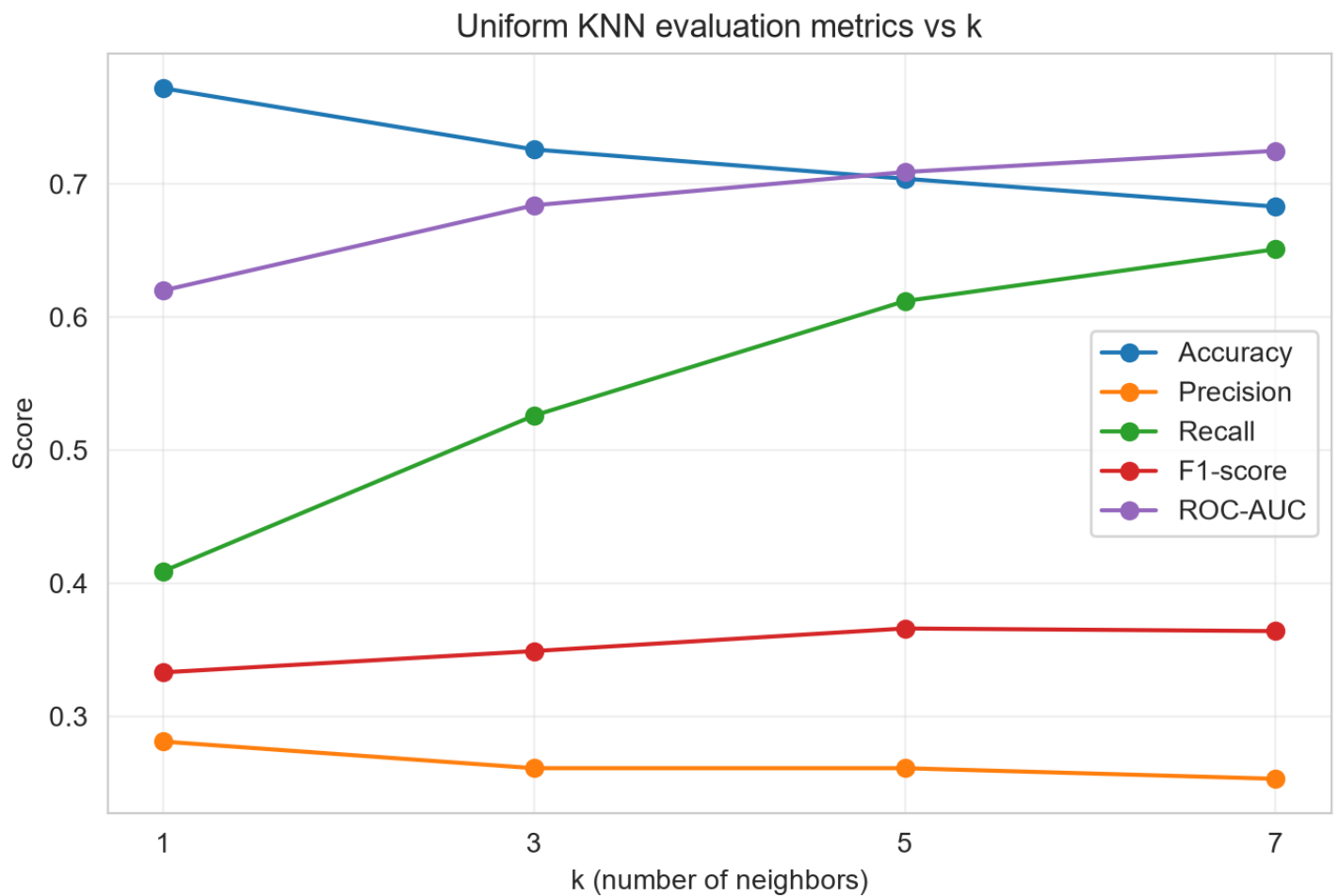
### 4. KNN performance across k values

| k | Model     | Accuracy | Precision | Recall | F1-score | ROC-AUC |
|---|-----------|----------|-----------|--------|----------|---------|
| 1 | KNN (k=1) | 0.772    | 0.281     | 0.409  | 0.333    | 0.62    |
| 3 | KNN (k=3) | 0.726    | 0.261     | 0.526  | 0.349    | 0.684   |
| 5 | KNN (k=5) | 0.704    | 0.261     | 0.612  | 0.366    | 0.709   |
| 7 | KNN (k=7) | 0.683    | 0.253     | 0.651  | 0.364    | 0.725   |

#### Weighting scheme :

#### Uniform (all neighbors counts equally)

| k | Model     | Accuracy | Precision | Recall | F1-score | ROC-AUC |
|---|-----------|----------|-----------|--------|----------|---------|
| 1 | KNN (k=1) | 0.772    | 0.281     | 0.409  | 0.333    | 0.620   |
| 3 | KNN (k=3) | 0.726    | 0.261     | 0.526  | 0.349    | 0.684   |
| 5 | KNN (k=5) | 0.704    | 0.261     | 0.612  | 0.366    | 0.709   |
| 7 | KNN (k=7) | 0.683    | 0.253     | 0.651  | 0.364    | 0.725   |

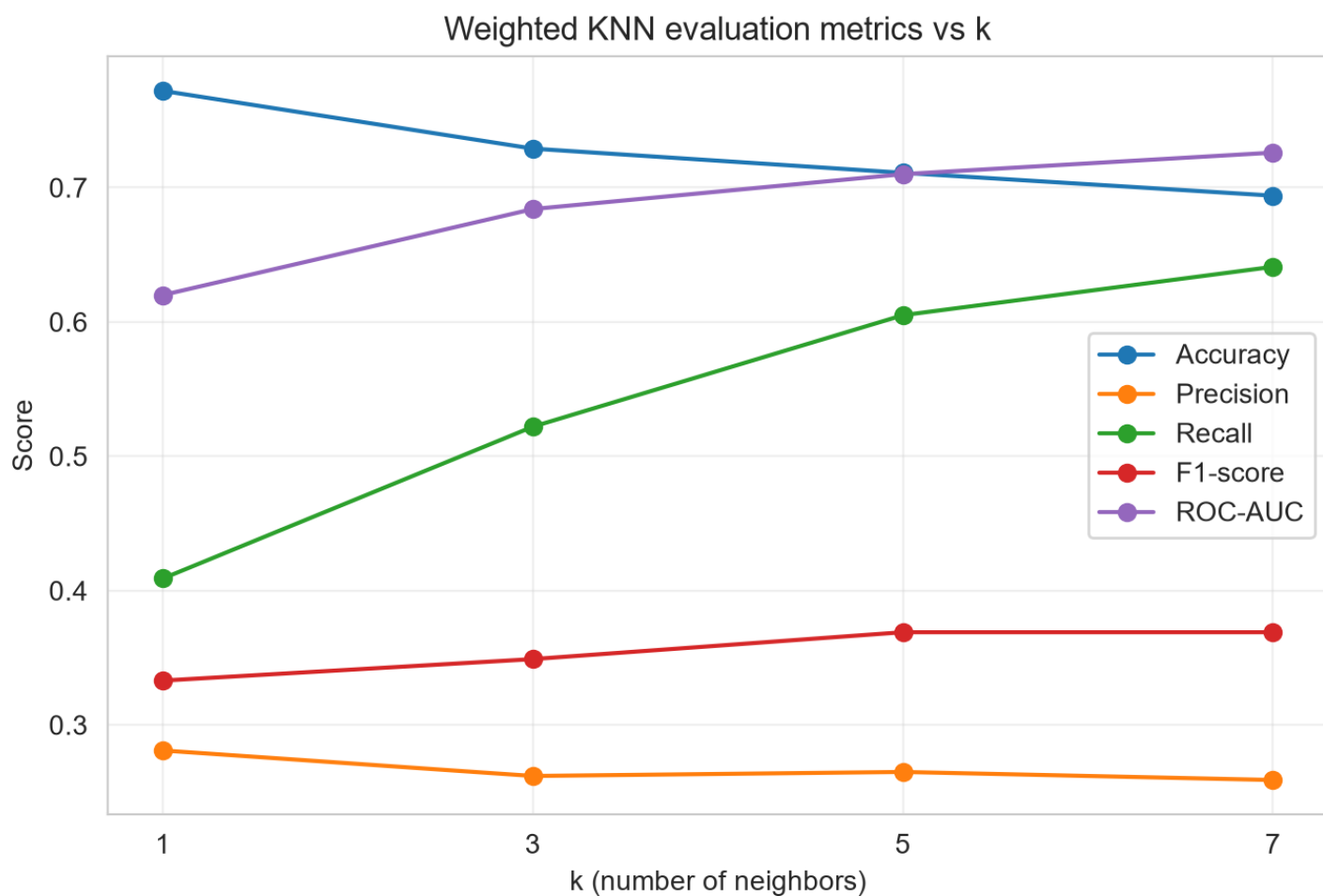


**Uniform KNN: best F1-score at k=5 (F1=0.366).**

### Distance –Weighted (Closer neighbors Count more)

| k | Model              | Accuracy | Precision | Recall | F1-score | ROC-AUC |
|---|--------------------|----------|-----------|--------|----------|---------|
| 1 | Weighted KNN (k=1) | 0.772    | 0.281     | 0.409  | 0.333    | 0.620   |
| 3 | Weighted KNN (k=3) | 0.729    | 0.262     | 0.522  | 0.349    | 0.684   |
| 5 | Weighted KNN (k=5) | 0.711    | 0.265     | 0.605  | 0.369    | 0.710   |
| 7 | Weighted KNN (k=7) | 0.694    | 0.259     | 0.641  | 0.369    | 0.726   |

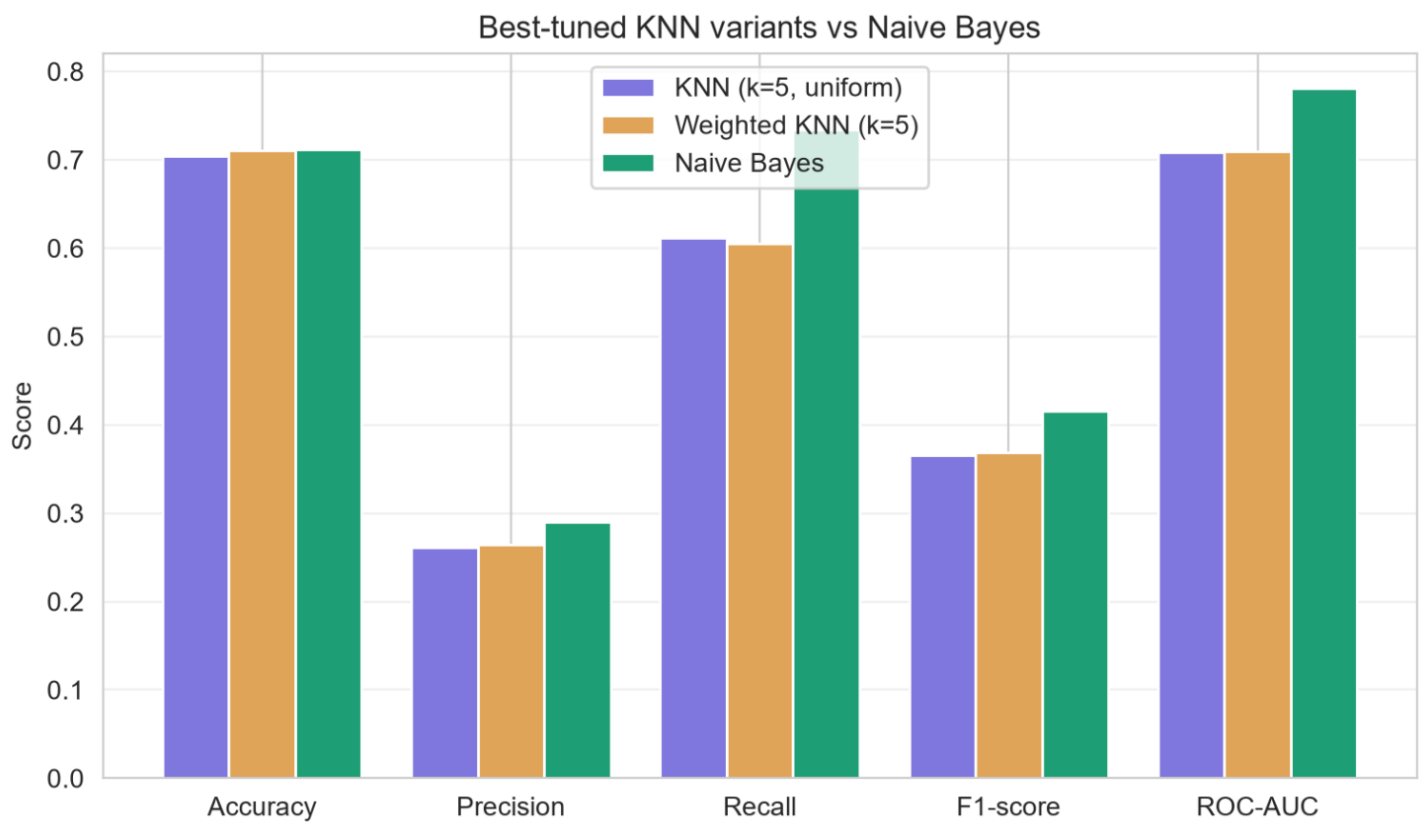




**Weighted KNN: best F1-score at k=5 (F1=0.369).**

## 5. Best Uniform KNN vs Best Weighted KNN vs Naive Bayes

| Model              | Accuracy | Precision | Recall | F1-score | ROC-AUC |
|--------------------|----------|-----------|--------|----------|---------|
| KNN (k=5, uniform) | 0.704    | 0.261     | 0.612  | 0.366    | 0.709   |
| Weighted KNN (k=5) | 0.711    | 0.265     | 0.605  | 0.369    | 0.710   |
| Naive Bayes        | 0.712    | 0.290     | 0.734  | 0.416    | 0.781   |



**Higher F1-score: Naive Bayes.**